

# **STAREHE GIRLS' 2025 PREDICTION**



## **POSSIBLE KCSE EXAM**



233/2

## **CHEMISTRY**

**PAPER 2 (THEORY)**

**TIME: 2 HOURS**

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

*(Kenya Certificate of Secondary Education)*

### **Instructions to candidates**

- ✓ Write your name and admission number in the spaces provided above.
- ✓ Answer **all** the questions in the spaces provided in the question paper.
- ✓ Non-programmable silent electronic calculators and KNEC mathematical tables may be used.
- ✓ All working **must** be clearly shown where necessary.
- ✓ Candidates should answer the questions in English.

1. The grid below represents part of the periodic table. Study it and answer the questions that follow. The letters do not represent the actual symbols of the elements.

|   |   |  |   |   |   |  |   |   |
|---|---|--|---|---|---|--|---|---|
|   |   |  |   |   |   |  |   |   |
|   |   |  |   | Z |   |  | B |   |
| R |   |  | Y |   | A |  |   | D |
|   |   |  |   |   |   |  |   |   |
|   | Q |  |   |   |   |  |   |   |
|   |   |  |   |   |   |  |   |   |

- a) Compare the melting points of R and Y.

(2 marks)

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- b) Basing on the structure and bonding, state the structure in;

(2 marks)

- i) Z.....
- ii) Oxide of Q.....

- a) Which element has the lowest ionization energy.

(1 mark)

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- b) State the nature of ;

(2 marks)

- i) Oxide of A.....
- ii) Oxide of Y.....

c) Choose the letter representing an element that;

**(2 marks)**

i) Is the most electronegative.....

ii) Exist as a monoatomic gas.....

d) Nitrogen falls in the same group as A.

i) Write the formula of the compound when A reacts with magnesium. **(1 mark)**

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ii) Write an equation for the reaction between the compound in f (i) above and water. **(1mark)**

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e) State the general property common to elements in the group containing D. Explain. **(2 marks)**

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2. Study the table below and use it to answer the questions that follow.

| Hydrocarbon                   | Boiling point (K) |
|-------------------------------|-------------------|
| CH <sub>4</sub>               | 112               |
| C <sub>2</sub> H <sub>6</sub> | 184               |

|             |     |
|-------------|-----|
| $C_3H_8$    | 231 |
| $C_4H_{10}$ | 273 |
| $C_5H_{12}$ | 309 |
| $C_6H_{14}$ | 342 |

**a)** These organic compounds belong to the same homologous series;

**i)** What is meant by the term homologous series?

**(1 mark)**

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**ii)** To which homologous series do the above hydrocarbons belong?

**(1 mark)**

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**ii)** Select one hydrocarbon that would be a liquid at room temperature, 298K. give a reason.

**(2 marks)**

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**iii)** What is the relationship between the boiling point and the relative molecular masses of the hydrocarbons in the table above? Explain.

**(2 marks)**

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**iv)** Describe one chemical test used to distinguish between  $C_2H_6$  and the third member of the homologous series with the general formula  $C_nH_{2n}$

**(2 marks)**

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**b)** Ethene gas can be prepared in the laboratory by heating ethanol with concentrated sulphuric (vi) acid.

i) Name the type of reaction which takes place.

**(1mark)**

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ii) State the temperature required in the reaction b i) above.

**(1**

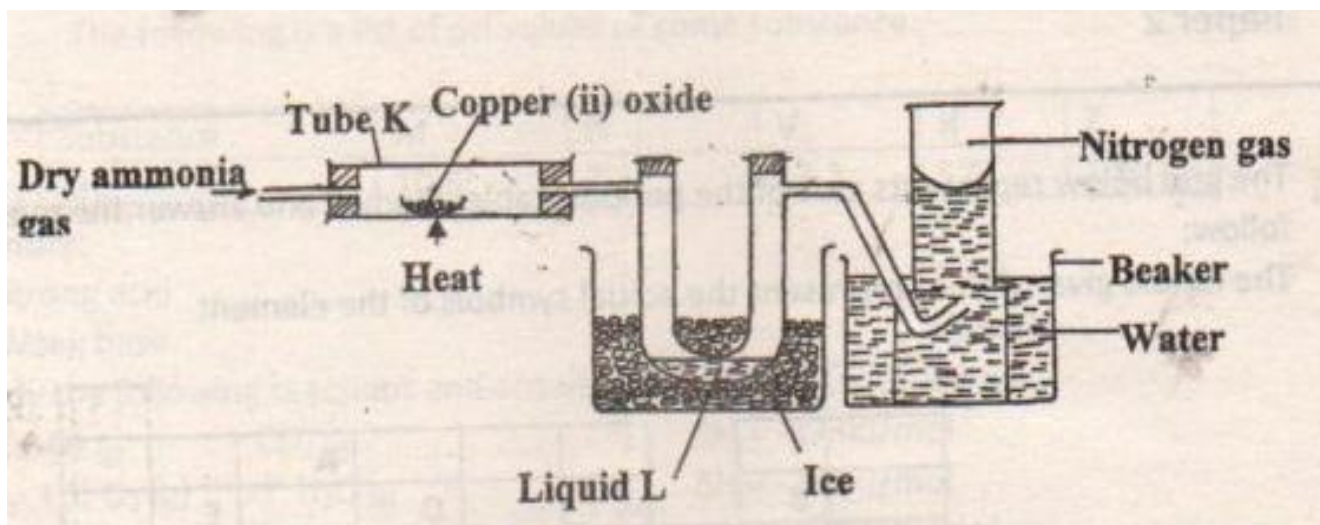
**mark)**

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iii) Write the chemical equation for the reaction which produces ethene gas. **(1 mark)**

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**3. a)** The diagram below shows the set up that can be used to obtain nitrogen gas in an experiment carried out by form three students.



i. How is the ammonia gas for this process dried?

(1mark)

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ii. Name liquid L.

(1 mark)

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iii. What observation would be made at tube K at the end of the experiment.

(1 mark)

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iv. Write an equation for the reaction that took place in the tube K.

(1 mark)

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v. At the end of the experiment the pH of water in the beaker was found to be 10.0 Explain.

(1mark)

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b) i) Ammonia decompose if sparked electrically. What would you expect to be the products of the decomposition?

(1 mark)

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ii) Describe the chemical test of ammonia.

(2 marks)

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c) Ammonia gas reacts with water according to the equation below.



i) Identify the species that acts as a base. Give a reason.

(2

marks)

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ii) What effect does addition of sodium hydroxide solution have on the position of the equilibrium?

Explain.

**(2 marks)**

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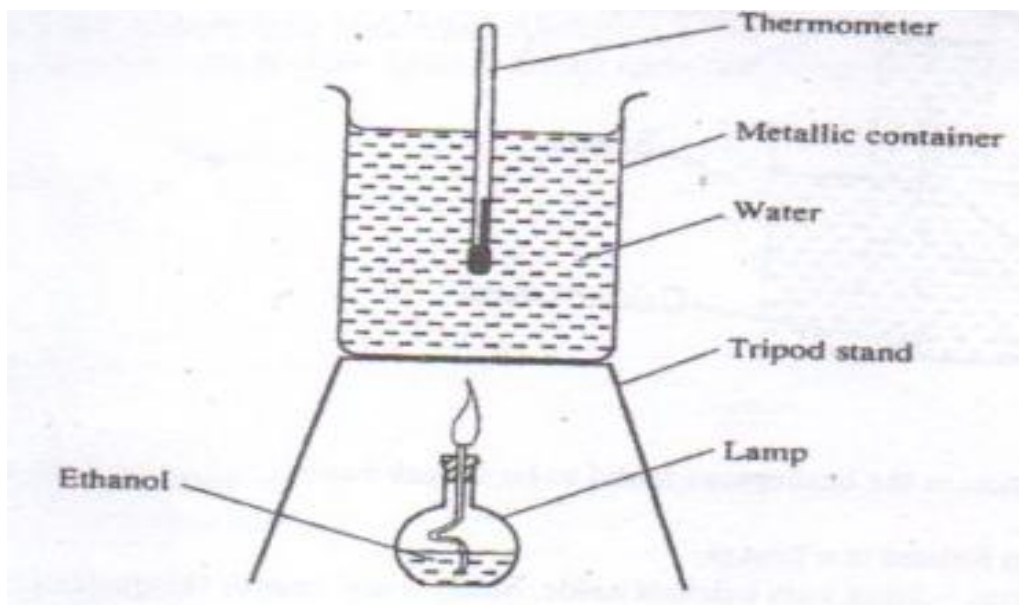
4. State two factors that should be considered when choosing fuel for cooking. **(2 marks)**

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(b) The diagram below represents a set –up that was used to determine the molar heat of combustion of ethanol



During the experiment, the data given below was recorded

|                                       |                     |
|---------------------------------------|---------------------|
| Volume of water                       | 450cm <sup>3</sup>  |
| Initial temperature of water          | 25 <sup>0</sup> C   |
| Final temperature of water            | 46.5 <sup>0</sup> C |
| Mass of ethanol + Lamp before burning | 125.5g              |
| Mass of ethanol + Lamp after burning  | 124.0g              |

**Calculate the:**

i. Heat evolved during the experiment (density of water=1g/cm<sup>3</sup>)

**(2marks)**

Specific heat capacity of water =4.2Jg<sup>-1</sup>K<sup>-1</sup>

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ii. Molar heat of combustion of ethanol (C=12.0, O=16.0, H=1.0)

**(2marks)**

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(c) Write the thermal chemical equation for the complete combustion of ethanol. (1mark)

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(d) The value of the molar heat of combustion of ethanol obtained (b) (ii) above is lower than the theoretical value. State two sources of error in the experiment. (2 marks)

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e) Define the term molar heat of combustion (1 mark)

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5. An impure solid of Copper (II) Carbonate weighing 10.8grams was placed in a beaker containing 50cm<sup>3</sup> of dilute Nitric (V) acid. The volume of Carbon (IV) Oxide evolved was recorded at 20 second interval in the table below.

|                                            |     |      |      |      |      |      |      |
|--------------------------------------------|-----|------|------|------|------|------|------|
| Time from start of reaction (se)           | 0   | 20   | 40   | 60   | 80   | 100  | 120  |
| Volume of CO <sub>2</sub> at S.T.P(litres) | 0.0 | 0.65 | 0.90 | 1.07 | 1.10 | 1.12 | 1.12 |

**a)** Write an equation for the reaction between copper (II) carbonate and nitric (V) acid. **(1mark)**

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**b)** Calculate the reaction between:

i. 20 seconds and 40 seconds intervals.

**(1 mark)**

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ii. 40 seconds and 60 seconds intervals.

**(1 mark)**

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**c)** Explain the difference in the reaction rates in (b) above.

**(2marks)**

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**d) Why was there no increase in volume of the gas after 100 seconds? (1 mark)**

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**e) How many moles of Carbon (IV) oxide were there in the maximum gas produced from this reaction? (M.G.V at S.T.P 22.4 litres). (2 marks)**

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**f) What mass of copper (II) carbonate that will have reacted with the acid after 100 seconds? (Cu=64, C=12, O=16) (2 marks)**

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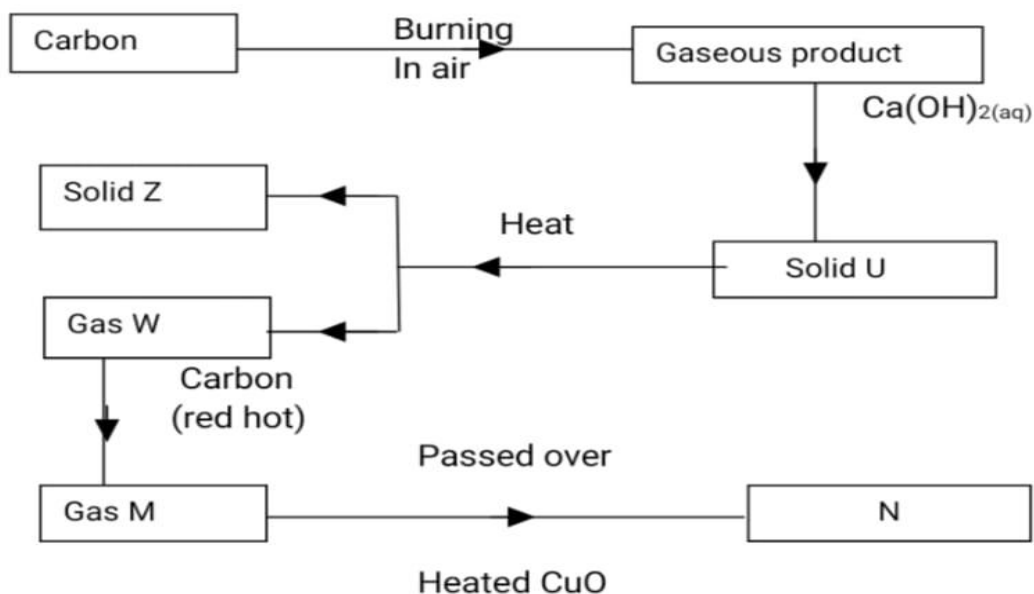
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**g) Calculate the original concentration of the nitric (V) acid in moles per litre. (2marks)**

6. a) The diagram below shows several reactions starting with carbon.



i. Identify the following U, Z, W, M  
(2marks)

U.....

Z.....

W.....

M.....

ii. State the observation made when gas M is passed over heated copper (II) oxide. (1 mark)

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iii. State one agricultural application of solid **Z**.

**(1mark)**

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iv. Write an equation for the formation of gas **M** from gas **W**.

**(1mark)**

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v. Explain the precaution to be taken in the reaction taking place in (iv) above

**(1mark)**

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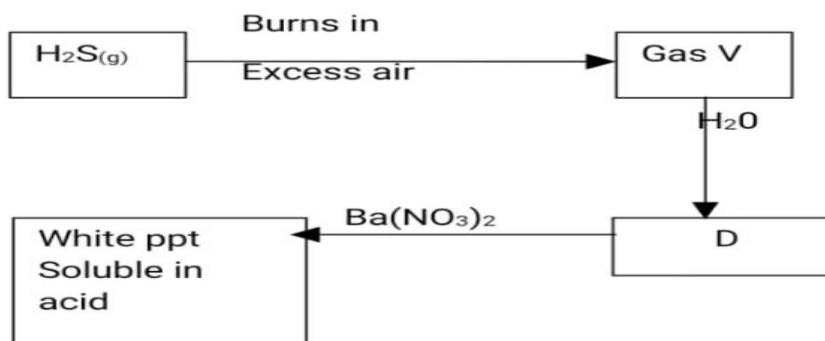
**(b) (i)** Compare the boiling point of Sulphuric (VI) acid to that of water. Explain **(2marks)**

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**(ii)** Study the flow chart below and answer the questions that follow



I. Name Gas V,D

(1mark)

V.....

D.....

II. Write an ionic equation for the formation of white precipitate.

(1mark)

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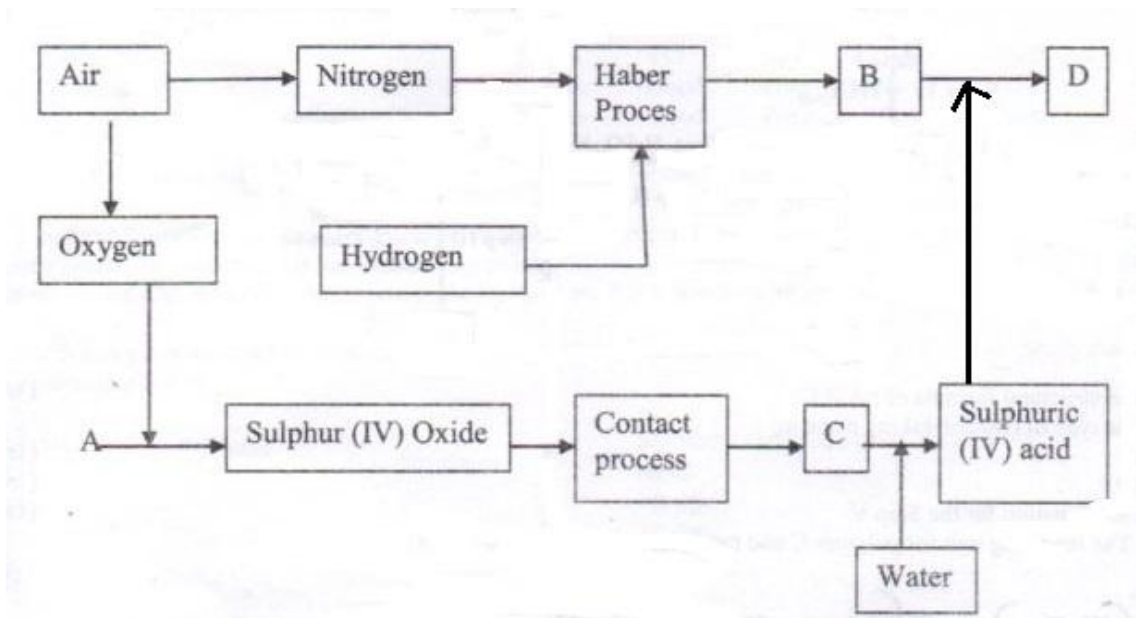
III State the observation made when dilute nitric acid is added to the white precipitate in II above.

(1 mark)

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7. The flow charts below illustrate two industrial processes, Haber process and the Contact process.



a) i. Give the name of the process by which air is separated into oxygen and nitrogen (1mark)

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ii. Apart from oxygen and nitrogen gases produced from process a (i) name one other gas produced.

(1 mark)

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b) Name substances represented by the letters A, B, C

(3marks)

A.....

B.....

C.....

**c) Name the catalyst used in:**

i) Haber process.....

**(1mark)**

ii) Contact Process.....

**(1mark)**

**d) Explain the role of the catalysts both the Haber process and the Contact processes** **(1mark)**

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**e) Write a chemical equation for the formation of compound B.**

**(1mark)**

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**f) Calculate the percentage by mass of the nitrogen present in compound D.**

**(1mark)**

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**g) State how compound C is formed.**

**(1mark)**

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